# v158th FW

# FLYING TRAINING SYLLABUS

**March 2026**

This syllabus revises and replaces all v158 FW F-35A Qualification or Upgrade Syllabi and previous versions of v158FWI 11-202V1. Direct any questions to v158 FW WG/CV at marcelinev158th@gmail.com.

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## SYLLABUS GUIDANCE

The v158th FW Flying Training Syllabus gives the requirements for Mission Qualification Training (MQT), Flight Lead training (FL), Mission Commander Training (MC), and Instructor Pilot Training (IP). Ground or flight training requirements or tasks may be modified at the discretion of the Squadron Commander. However, as a minimum, the MQT program in this syllabus will be followed as written for pilots who are assigned to the unit. The Operations Group Commander may alter this program to accommodate the background, experience, and demonstrated capabilities or skillsets of the individual pilots. Any changes from this syllabus must be approved by the Operations Group Commander.

The intent of the v158th FW Flying Training Syllabus is to provide basic guidance for qualification and upgrade requirements. The v158th FW has the freedom to build on or expand the outlines within the syllabus to meet the changing threats and tactics. The v158th FW Flying Training Syllabus will be reviewed at the discretion of the Squadron Commander, Operations Group Commander, Wing Vice Commander, or Wing Commander. Any changes that add or delete specific training or requirements will be approved, added, and deleted by the Operations Group Commander.

## SCHEDULING GUIDANCE

When scheduling sorties, the requirements stated in this syllabus are desired numbers which will provide the upgrading pilot with an optimal training environment. If these numbers are unachievable, whether in planning phase or execution, the IP of Record will determine the effectiveness and perceived training value of the sortie when applicable.

## TRAINING DOCUMENTATION

Gradebooks will be maintained on all students during and after their time in an upgrade program. Gradebooks will contain content including but not limited to: approved upgrade program plan & flow, syllabus events log, flying training summary, gradesheets, gradebook review sheet, and any letters or waivers addressing changes to the normal upgrade training flow. MQT students will also have a theater indoctrination folder that must be completed prior to any flying events.

## GRADESHEETS

Gradesheets will be accomplished immediately after the completion of upgrade training and made available for the next IP's review. Gradesheets should reflect the student's progression and should be addressed and tailored for the next IP, rather than the student. Missions deemed ineffective by the IP of Record will be graded as "NG" (No Grade) with an explanation on the respective gradesheet. Missions that are ineffective for the student's performance and growth will

be graded as a “1” or less overall with an explanation in the remarks section and the following mission will be flown as an “X” ride (i.e., MQT-3X).

Alternate mission profiles are limited to maneuvers, mission events and mission profiles that have been flown to a 2-level or higher on a previous syllabus training sortie. Supervisors will evaluate the need for the removal of a student from any upgrade program where several X rides are required.

Grades will be assigned to the overall mission and to each task within the mission. Overall mission grades are as follows:

U Unsatisfactory performance due to demonstrated lack of ability. Overall performance indicates that corrective action is required.

F Marginal performance. Does not meet standards for all tasks requiring proficiency, or does not warrant progression.

G Satisfactory performance. Performs in essentially the correct manner. Recognizes and corrects errors. Performance meets standards for all tasks requiring proficiency.

E Excellent performance. Performs correctly, efficiently, skillfully, and without Hesitation. Performance exceeds standards.

Any grade that is not a “2” or above requires amplifying remarks on the gradesheet. The overall mission grade reflects the IP of Record’s evaluation of overall performance for that stage of the upgrade program. Individual task grades are as follows:

NG Performance was not observed, task was not performed, or no evidence.

D Performance was unsafe. One element marked “dangerous” on the gradesheet will be recorded as an overall grade of “0.”

U Performance indicated a lack of ability or knowledge required to accomplish the event.

F Performance was safe, but indicated limited proficiency. Makes errors of omission or commission.

G Performance is essentialized and standard. Recognizes and corrects errors.

E Performance is correct, efficient, skillful, and without hesitation.

Individual tasks that are graded as a “F” do not require an overall mission grade of “F” if proficiency was not required to be demonstrated on that mission. Any specific mission events that are not accomplished will be graded as a “U” and annotated on the gradesheet as incomplete. The event will also be left incomplete on the Syllabus Events log. All mission tasks will be accomplished before the end of any upgrade program. Otherwise, the final mission will be annotated as incomplete and the task will be accomplished as soon as possible.

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# Chapter 1

# Mission Qualification Training

## Section A. General Instructions

### 1-1. General Instructions for All Training

a. Instructors may adjust mission tasks and profiles around individual student needs and manpower constraints, but must ensure all students receive all syllabi and mission objective material.
b. Mission configurations are at the discretion of the instructor to ensure mission completion, unless specifically stated otherwise in the syllabus notes.
c. Appropriate training record documentation is required for all missions and instructions.
d. Schedule students and instructors for both a 30-minute briefing and 30-minute debriefing for all MQT missions.

### 1-2. Mission Accomplishment

a. Vehicle Systems Training (VST): Academic portion. Accomplished outside of the airborne environment, available within resources provided to all upgrading students.
b. Mission Systems Training (MST): Academic portion. WST missions may be conducted during MQT missions at the discretion of the IP of Record, but will typically be accomplished outside of the airborne environment.
c. Mission Qualification Training (MQT): Flight phase. Primary objective and upgrading curriculum to be provided to students.

**Section B. Classes**

**1-3. Class Chart:** VST, MST, MQT

| *Class* | *Description* | *Prerequisites* |
|---|---|---|
| TI ACAD | Theater Indoctrination & FFAM/LAO | MST |
| VST-1 | Aircraft & Flight Controls Familiarization | N/A |
| VST-2 | Cockpit Familiarization | VST-1 |
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## 1-4. Vehicle Systems Training (VST)

### a. VST-1: Aircraft & Flight Controls Familiarization

i. Mission Objectives:
   1. Introduce:
      a. Aircraft Description.
      b. Aircraft Design Overview.
      c. Aircraft Variants & Differences.
      d. Engine.
      e. HOTAS Overview.

ii. Mission Tasks:
   1. Directed self-study of assigned aircraft systems material.

### b. VST-2: Cockpit Familiarization

i. Mission Objectives:
   1. Introduce:
      a. Full Cockpit Overview.
      b. Ejection Seat.
      c. Left Console.
      d. Throttle.
      e. Left Auxiliary Console.
      f. Panoramic Cockpit Display.
      g. Center Console.
      h. Rudder & Brake Pedals.
      i. Right Auxiliary Console
      j. Stick.
      k. Right Console.

ii. Mission Tasks:
   1. Directed self-study of assigned aircraft systems material.

### c. VST-3: Cockpit Display System I

i. Mission Objectives:
   1. Introduce:
      a. Detailed Panoramic Cockpit Display Overview.
      b. Status Bar.
      c. Function Access Buttons.
      d. Portals & Subportals.
      e. Pages.
      f. Integrated Caution, Advisory, and Warning System.

ii. Mission Tasks:
   1. Directed self-study of assigned aircraft systems material.

**d. VST-4: Cockpit Display System II**

i. Mission Objectives:

1. Introduce:

a. Detailed Helmet Mounted Display System Overview.
b. Helmet Anatomy.
c. Symbology.
d. On vs. Off Axis.
e. Standby Flight Display.
f. Display of Interest.
g. Portal of Interest.

ii. Mission Tasks:

1. Directed self-study of assigned aircraft systems material.

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## 1-5. Mission Systems Training (MST)

**a. MST-1: Navigation & Communication:**

i. Mission Objectives:

1. Introduce:

a. Navigation General Information.
b. Inertial Navigation System / Global Positioning System.
c. Electronic Flight Instruments.
d. Course Deviation Indicator.
e. Nav Sources.
f. Navigation Data Entry.
g. Waypoints.
h. GCAS.
i. Instrument Landing System.
j. Communication System Overview.
k. CNI Summary.
l. Voice Communications.
m. MADL/L16.
n. VMF.
o. Transponder Overview.
p. Identify Friend or Foe.

ii. Mission Tasks:

1. Directed self-study of assigned aircraft systems material.

**b. MST-2: TSD & Radar**

i. Mission Objectives:

1. Introduce:

a. Tactical Situation Display Overview & Controls.
b. Horizontal Situation Display Overview & Controls.
c. Radar Overview.
d. Search Volumes Summary.
e. Air-to-Surface Radar.

ii. Mission Tasks:

1. Directed self-study of assigned aircraft systems material.

**c. MST-3: DAS & EOTS**

i. Mission Objectives:

1. Introduce:

a. Electro-Optical Distributed Aperture System Summary.
b. Electro-Optical Targeting System Summary.
c. Electro-Optical Controls.

ii. Mission Tasks:

1. Directed self-study of assigned aircraft systems material.

**d. MST-4: Weapons Employment**

i. Mission Objectives:

1. Introduce:

a. Stores Management System Overview.
b. Stores Management System Controls.
c. Stores Management System Usage.
d. Air-to-Air Weapons Summary.
e. Detailed Air-to-Air Weapons Employment Overview.
f. Detailed AA Gun Employment Overview.
g. Air-to-Surface Weapons Summary.
h. Detailed Air-to-Surface Weapons Employment Overview.
i. Detailed AS Gun Employment Overview.

ii. Mission Tasks:

1. Directed self-study of assigned aircraft systems material.

## 1-6. Mission Qualification Training (MQT)

### a. MQT-1: Offensive BFM

i. Mission Objectives:
   1. Practice local area procedures.
   2. Demonstrate Proficiency in:
      a. Standard Operating Procedures.
      b. Takeoff and Landing.
      c. Offensive BFM.
      d. Daytime AAR if avail.

ii. Mission Tasks:
   1. Classroom.
   2. Briefing.
   3. Takeoff.
   4. G-awareness Exercise.
   5. DAAR, if avail.
   6. Offensive BFM for UP at 3K’, 6K’, 9K’.
   7. Landing.
   8. Debrief.

### b. MQT-2: Defensive BFM

i. Mission Objectives:
   1. Practice local area procedures.
   2. Demonstrate Proficiency in:
      a. Standard Operating Procedures.
      b. Takeoff and Landing.
      c. Defensive BFM.
      d. Daytime AAR if avail.

ii. Mission Tasks:
   1. Classroom.
   2. Briefing.
   3. Takeoff.
   4. G-awareness exercise.
   5. Defensive BFM for UP at 3K`, 6K`, 9K`.
   6. Landing.
   7. Debrief.

**c. MQT-3: High-Aspect BFM**

i. Mission Objectives:

1. Practice local area procedures.
2. Demonstrate proficiency in High-Aspect BFM.

ii. Mission Tasks:

1. Classroom.
2. Briefing.
3. Takeoff.
4. High-Aspect BFM in Butterfly Setup.
5. Instrument Trail Recovery
6. Landing.
7. Debrief.

**d. MQT-4: BVR TI Launch & Leave**

i. Mission Objectives:

1. Demonstrate proficiency in local area procedures.
2. Demonstrate proficiency in L&L employment.
3. Demonstrate proficiency in normal and emergency instrument procedures at Burlington International Airport.

ii. Mission Tasks:

1. Classroom.
2. Briefing.
3. Takeoff.
4. Radar based TI.
5. Employment of TI L&L.
6. ILS approach.
7. TACAN approach.
8. Emergency divert.
9. Landing.
10. Debrief.

**e. MQT-5: Element TI & ACM:**

i. Mission Objectives:

1. Demonstrate proficiency in Tactical Intercept.
2. Demonstrate proficiency in Perch and All-Aspect ACM.

ii. Mission Tasks:

1. Classroom.
2. Briefing.
3. Takeoff.
4. Element Tactical Intercept.
5. All-Aspect ACM Setup.

6. Perch ACM Setup.
7. Landing.
8. Debrief.

**f. MQT-6: CAS**

i. Mission Objectives:
1. Demonstrate proficiency in local area procedures.
2. Demonstrate proficiency in TAC communication.
3. Demonstrate proficiency in Killbox operations.
4. Introduce precision air-to-surface weapons employment.
5. Introduce air-to-surface gun employment.

ii. Mission Tasks:
1. Classroom.
2. Briefing.
3. Takeoff.
4. Navigation.
5. TAC communication.
   a. 9-line tasking.
   b. Employment.
6. Weapons employment
   a. LGB or JDAM.
   b. Cannon.
7. Landing.
8. Debrief.

**g. MQT-7: 2vX TI**

i. Mission Objectives:
1. Demonstrate proficiency in local area procedures.
2. Demonstrate proficiency in Launch & Decide Tactical Intercepts.

ii. Mission Tasks:
1. Classroom.
2. Briefing.
3. Takeoff.
4. Radar based TI.
5. Employment of TI L&D.
6. Landing.
7. Debrief.

**h. MQT-8: 2vX ACM**

i. Mission Objectives:
   1. Demonstrate proficiency in local area procedures.
   2. Demonstrate proficiency in Air Combat Maneuvering.

ii. Mission Tasks:
   1. Classroom.
   2. Briefing.
   3. Takeoff.
   4. All-Aspect ACM.
   5. Landing
   6. Debrief.

**i. MQT-9: SCAR & Air Interdiction**

i. Mission Objectives:
   1. Demonstrate proficiency in local area procedures.
   2. Demonstrate proficiency in precision air-to-surface engagements.
   3. Demonstrate proficiency in LFE cooperation and presence.

ii. Mission Tasks:
   1. Classroom.
   2. Briefing.
   3. Takeoff.
   4. LFE Navigation.
   5. Precision Air Interdiction.
   6. Landing.
   7. Debrief.

**j. MQT-10: Night Flying**

i. Mission Objectives:
   1. Introduce usage of NVCS.
   2. Practice local area procedures under night conditions.

ii. Mission Tasks:
   1. Classroom.
   2. Briefing.
   3. NVCS familiarization.
   4. Takeoff.
   5. Instrument procedures under night conditions.
   6. 2-ship maneuvering under night conditions.
   7. Precision approaches under night conditions.
   8. Non-precision approaches under night conditions.
   9. Landing.
   10. Debrief.